

YEAR 9 ENGINEERING

Water Tower

STUDENT WORKBOOK



A light structure. A clear explanation.

Read the ideas. Develop your design. Show what the test reveals.

NAME _____ CLASS _____

Illustrative model, not a dimensioned plan | A4 | 59 pages | Duplex long edge

How to use this workbook

Begin with safety and structural concepts, then use those ideas to plan your tower. Record evidence as you work: a dated sketch, a labelled photograph or a careful observation explains a decision better than a memory written at the end. Your teacher guides demonstrations, approves practical work and confirms the test arrangements.

This is a formative learning task. Use feedback to improve your understanding and work; it is not graded and does not contribute to your final mark. Your own measured results belong in the blank fields, while examples are clearly labelled.

Learning route

Start here	Project brief and success criteria
Toolkit 1	WHS and risk control
Toolkit 2	Structural concepts
Toolkit 3	Design and planning
Toolkit 4	Build, test and evaluate
Project folio	Sketches, build evidence, data and reflection

Some lessons include teacher-led demonstrations, videos or a bridge simulator. Read the print explanation first, then use the activity spaces to record what the demonstration shows. Keep OnGuard evidence and separate teacher-issued safety sheets with your project record.

Teacher-confirmed folio date: _____

Related reflection prompts use the same labelled drawing, table or record. Address each prompt in that evidence and identify any continuation sheet. Keep predictions separate from observations and leave teacher approval fields for your teacher.

The design challenge

Build a tower that holds heavy weight safely and efficiently using the supplied material.

What your tower must do

Height	At least 300 mm tall.
Mass	No more than 60 g.
Load	Must hold at least 25 kg.
Time	Must hold that load for at least 5 seconds.
Top	Top must support a 300 mm square board during testing.

Supplied materials and testing items

Structural members are 4 mm square cross-section plywood. During testing, a 300 mm square board sits on top and cast iron weights are placed on it. Design around the supplied members and the actual teacher-approved test setup.

Strong designs combine triangles and bracing, neat joints, a stable base and a clear path from the top support through the frame to the base. A tower that wobbles or transfers load through a weak joint can fail even when individual members appear strong.

Teacher-confirmed team members: _____

Underline each limit above. Beside it, write how you will measure or check it.

Criteria become evidence

A requirement tells you what to achieve; evidence shows whether you achieved it. Decide how to record each check before building. A photograph can show a feature, but a claim about mass or supported load needs a measurement. Keep a unit beside every number.

From claim to check

Requirement	Evidence to collect	Result
Minimum height	Overall height in mm	
Maximum mass	Tower mass in g	
Target load and time	Approved-test load in kg and time in s	
Top-board support	Annotated sketch/photo and pre-test check	

Your folio explains how you arrived at the design: ideas, drawings, hazards and controls, materials and tools, sequence, dated build evidence, data and evaluation. Record changes honestly. A design that needs improvement can still provide valuable learning when you explain the evidence carefully.

Which statement matches the height requirement?

- A. The tower must be at least 300 mm tall.
- B. The tower must be no more than 300 mm tall.
- C. The tower must be exactly 300 mm tall.

Choose one design feature you want to check. Name the evidence that would show whether it worked.

Safe work starts before a tool

WHS protects people, equipment and the workshop. It begins before the tool is switched on: understand the hazard, organise the work area and check controls. Training helps you recognise danger, but it does not replace teacher permission for the tool and lesson. Only use tools you have been trained and approved to use.

Wear safety glasses in the workshop and other protection required for the task. Tie long hair back; remove jewellery and cords; do not wear loose clothing around rotating tools. Loose heads, cracked handles, damaged cords, missing guards or chipped cutting edges are reasons to stop and report the problem.

The line of fire is the path a tool or material may take if it slips, breaks or moves unexpectedly. Clamping keeps small work secure and helps keep the supporting hand away from that path. Before cutting or drilling, think about tool movement and what might move if the setup fails.

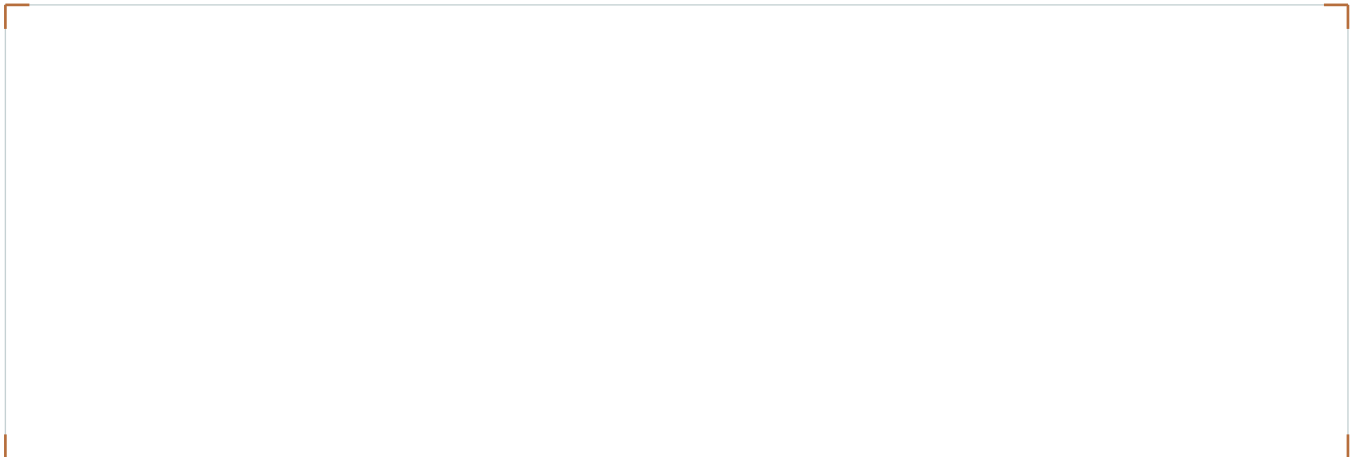
OnGuard completion

Complete the required OnGuard modules (hand tools, power tools/machinery).

Capture proof of completion (certificate, screenshot or teacher record).

Completion evidence / teacher record: _____

Sketch the teacher-demonstrated setup for the approved task. Mark a possible line of fire and show where the work is securely supported or clamped. In 2–3 lines, explain your support and hand-position choices. Use the actual SOP and setup confirmed by your teacher.



Inspect, support and control

Hand tools need a planned setup. Measuring and marking tools have points or springing blades: keep divider points closed and control a tape blade as it returns. Saws and knives need supported work and a cutting direction away from hands and body. Cover or retract sharp blades when finished.

Use a bench hook or vice where appropriate for chiselling or planing and keep the non-working hand out of the cutting path. Check hammer and mallet heads and handles before striking. A clamp should grip firmly without crushing material; clean jaws and a smooth screw help prevent slipping. Use the correct screwdriver or bit with controlled pressure.

Carry sharp tools with edges down or covered and return them to their racks; do not leave tools over bench edges. Respect shared equipment, clean the work area and report unsafe behaviour immediately. Never attempt repairs yourself.

Power tools add rotating-part, electrical, noise and heat hazards. Plan a stable stance and a cord route that will not trip someone. One operator uses a power tool unless the teacher sets up a supervised two-person task. Wait for a complete stop before putting it down and isolate power before changing bits or cleaning.

Inspection reveals damage to a tool cord. What is the best next action?

- A. Stop and report it to the teacher.
- B. Continue because the OnGuard test is complete.
- C. Try to repair it yourself.

Photo evidence

Take or use photos (teacher-approved) showing: correct PPE in use; work clamped for drilling; glue gun on its stand with drip tray.

Use these as part of your folio or reflection.

Caption one approved photo: what control is visible, and what hazard does it reduce?

Control the hazard; read the sign

Begin with the most effective control. Removing a trip hazard eliminates it; changing to a safer process substitutes a lower-risk approach. Barriers and equipment design reduce exposure before procedures and PPE support the setup. PPE protects the person but does not make a poor setup safe.

The original six-level hierarchy

1 Elimination	Physically remove the hazard
2 Substitution	Replace with a safer option
3 Isolation	Separate people from the hazard
4 Engineering	Improve equipment or processes
5 Administrative	Training, procedures and signage
6 PPE	Protect the worker

Toolkit 1 groups isolation and barriers with engineering controls in a five-level summary; the original reference separates them. Both place elimination first and PPE last. Explain what the control does, rather than relying only on a label.

Mandatory signs use a blue circle; prohibition signs a red circle and slash; warning signs a yellow triangle. Green rectangles identify emergency information. Red rectangles or squares locate fire equipment, used only when trained or instructed. Obey signs and report missing or unclear ones.

Workshop safety sign

Design a safety sign for a real workshop hazard (e.g. eye protection, hot surfaces, no running).

Use the correct category (mandatory, prohibition, warning) and include text and symbol.

Draft the safe method

A SWMS connects the actual work sequence to hazards and controls. Describe each step clearly enough that the risk can be recognised, then identify who uses the control and when. Check controls with your teacher before practical work. A draft is learning evidence and does not replace the school-approved procedure.

Using the Risk Management Plan template (PDF): identify hazards for hand tools, the corded drill and the hot glue gun.

Apply the hierarchy of control to each hazard.

Write clear, step-by-step safe work methods for tower tasks.

Work step	Hazard	Control and hierarchy level	Who / when

The linked risk plan is a historical drill-press example. Its eye-injury row shows a control assigned to an operator and time; its class, room and date do not describe this project. Use the current teacher-approved plan and matrix. Review controls when an activity changes or an incident shows a weakness.

Teacher feedback / changes to make: _____

Google Classroom check-in

Complete the Work, Health & Safety starter activity.

Finish the Safety & OnGuard module before practical work.

Submit your responses to the WHS & Risk Management task alongside your SWMS draft.

Corded drill: a controlled setup

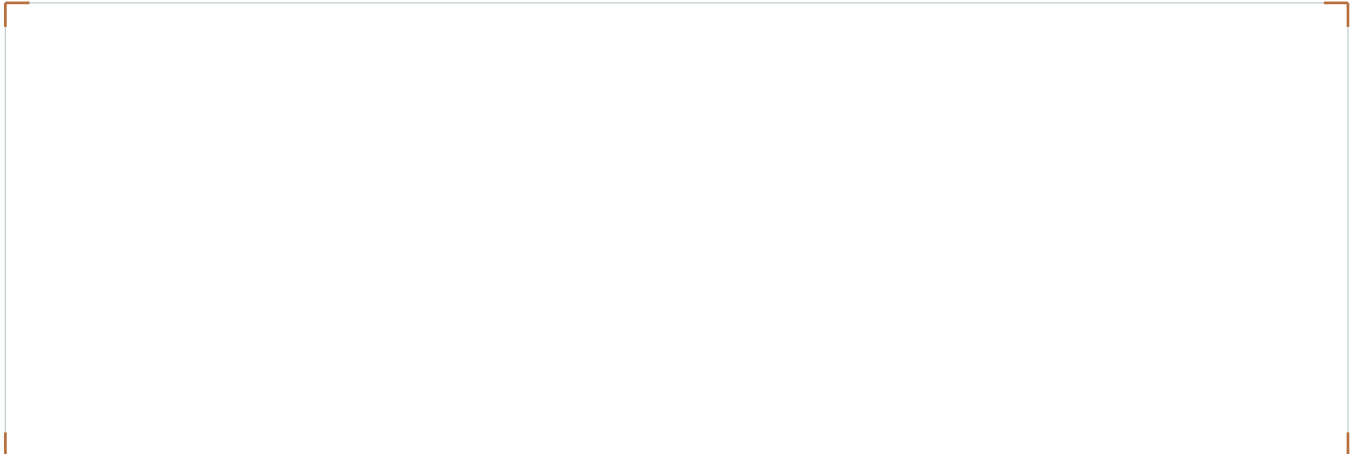
A corded drill can throw chips, catch hair or clothing, spin unsecured work, kick back or expose a damaged electrical connection. The setup should address those hazards before the bit rotates. Wear safety glasses and the task protection needed for noise, dust and footwear; tie long hair back, remove jewellery and cords, and do not wear loose clothing around rotating tools.

Check the cord, plug, casing and vents. The bit must be correct for the material, straight, sharp and secured in the chuck. Remove the chuck key before starting. Clamp the work and keep hands away from the area behind or beneath the hole, where the bit may emerge. Route the cord so it cannot pull the tool or trip someone.

The teacher demonstrates settings for the actual task. The source distinguishes lower speed for metal or large bits from higher speed for timber or small bits, and reserves impact mode for masonry. These are general principles, not a numerical setting for your tower. Use two hands, including a side handle if fitted, begin slowly and let the drill work without forcing it.

Keep fingers clear of rotating parts and remember the bit can be hot after use. If the tool vibrates, smells burnt or sounds unusual, stop and inform the teacher; isolate it if safe. Wait for a complete stop before putting it down and unplug before changing a bit or cleaning. When cool, clean and store the drill with its cord loosely coiled and off the floor.

Sketch the teacher-demonstrated corded-drill setup. Label three controls and the hazard each addresses. In 2–3 lines, explain how those controls reduce risk. Use the actual SOP and setup confirmed by your teacher.



Hot glue: manage heat and contact

A glue gun adds a heat hazard to joining. The nozzle and melted glue can burn skin; drips, a damaged cord and an unstable resting position also create risk. Prepare a clear bench with a drip tray or scrap under the nozzle. Use the stand whenever the gun is resting.



Illustrative glued-member contact. Inspect the fit and alignment; the teacher confirms the actual joint and adhesive.

Inspect the cord and plug, check the stand is stable and use correct dry glue sticks for the model. Heat the gun on its stand and never test the nozzle by touching it. Hold the insulated handle, use gentle trigger pressure and keep the other hand outside the glue path. Follow the teacher demonstration for joining the actual parts.

Do not over-apply glue. Press parts together lightly and adjust before the glue sets, while keeping clear of hot glue. Return the gun to its stand between uses, keep cords outside walkways and never pull a partly melted stick out through the back. If glue contacts skin, do not wipe it; the source directs cooling under running water and immediately telling the teacher for first aid.

Switch off and unplug when it will not be used for a while. Leave it on the stand to cool rather than laying it on its side. Remove glue strings only when cool. The permitted adhesive, joint detail and cure time for your tower are confirmed for the supplied materials; glue-gun safety training does not decide the structural joint.

Write a short safe setup–use–pack-up sequence. Include the stand, hand position and cooling stage.

Prepare a safe test

A load test stores energy in raised masses and strained members. Control the risks from falling weights, collapse and unexpected movement. Review the SWMS, organise PPE, clear the fall zone and make sure the supervising teacher knows the setup. Nobody places hands beneath a load.

Before loading, check that the work area is clear, base and platform are stable, fixtures are set and the agreed stop protocol is understood. The original project uses a top board and cast iron weights. Your teacher confirms the rig, increments, permitted observations and stopping conditions. A website example is not permission to improvise a rig or load to failure.

First response: the source overview

Toolkit 1 introduces DRABC: Danger, Response, Airway, Breathing and CPR/Circulation. First decide whether it is safe to approach. Get the teacher or first aider and follow their instructions; airway assistance and CPR in the source are limited to trained people. Report cuts, burns and eye injuries immediately. Use this overview to recognise the need for help; the school’s trained response governs actual first aid.

Before the class test, record the teacher-confirmed controls for people, load placement and the stop signal.

People / safe zone	
Load placement / fixtures	
Stop signal / supervisor	

WHS check: training and setup

Answer in your own words before checking the model response with your teacher. Explain the reason for a control, not only its name. These questions practise safe decisions.

Why do you need teacher permission and OnGuard completion before using power tools?

What is the minimum PPE expected in the workshop?

What does “line of fire” mean when using hand tools?

WHS check: control and reporting

What is the first control to consider in the hierarchy of control?

What should you do if a tool vibrates, smells burnt or sounds wrong?

Why must small work be clamped instead of held by hand when drilling or cutting?

WHS check: heat and first response

What is the correct response if hot glue contacts skin?

In DRABC, what is the first step and why?

Name one control you now understand better. Explain how you will apply it during the tower project.

Mass is not force

Mass tells you how much matter is present and is measured in kilograms. Weight is the force due to gravity and is measured in newtons. The distinction matters when you draw force arrows or compare performance: mass becomes a load because gravity acts on it. Write the unit with each value.

SI units used in this unit

Mass	kg; convert g to kg when the calculation needs kg
Force	N or kN
Length	m or mm
Area and volume	m ² and m ³

Worked source example

Weight (N) = mass (kg) × gravitational acceleration (m/s²). Toolkit 2 uses 9.8 m/s² and allows 10 m/s² for simple class calculations. A 5 kg mass therefore applies 5 × 9.8 = 49 N, approximately 50 N when gravity is rounded to 10. This is a calculation example, not a new tower target.

Keep the rounding convention consistent. Do not write “kg of force”. Record test mass in the brief’s unit, then convert when a force calculation is needed. A measurement and its converted force describe the same load in different ways.

Which unit should label a force arrow on a sketch?

- A. kg
- B. N
- C. mm

Using 9.8 m/s², calculate the force associated with the brief’s 25 kg target mass. Show the equation and units.

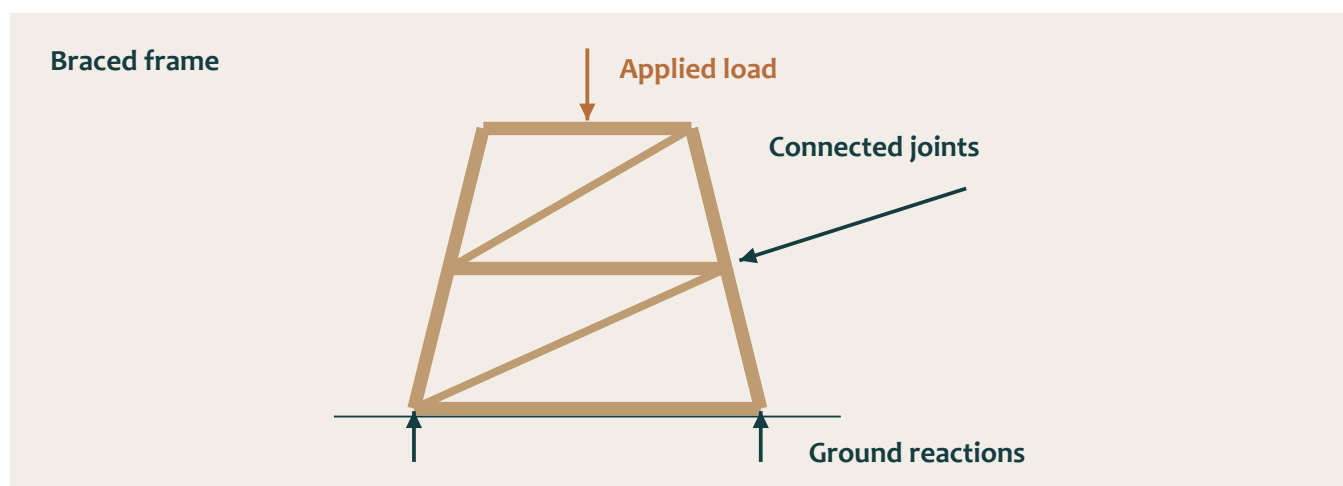
Follow the load to the ground

Start with the load: where it acts, how it is applied and how it reaches the supports. Dead load is permanent self-weight. Live load changes, such as water level or added test weights. Wind is an environmental load that can push sideways and challenge stability.

A point load acts at one location; a distributed load spreads over a length or area. The loading board transfers the test load to supporting members. Members and joints then carry it through the frame to the base. Trace that continuous route with arrows rather than simply writing “strong” beside the drawing.

The ground provides reactions. In the source’s ideal symmetrical tower, a centred point load and four identical supports share load approximately equally. This assumes symmetrical geometry, support behaviour and loading. Moving the load or making the frame uneven can increase the load on some supports and reduce it on others.

Sketch arrows for the applied load and reactions. Mark support behaviour where known; pinned, roller and fixed supports restrain movement differently. Do not claim a particular restraint for the rig until its setup is confirmed.



Conceptual front-view frame. External load and ground reactions are shown; this is not a dimensioned project drawing.

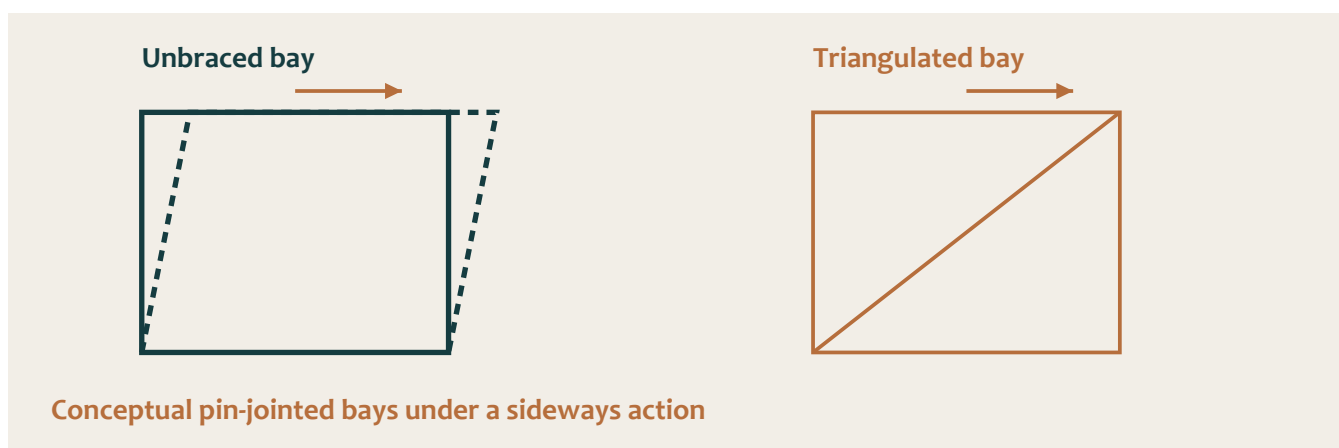
On the concept drawing, show a continuous load path from the top support to the ground. Mark a joint that transfers force between members.

Triangles resist racking

A four-sided frame can rack: its rectangle changes towards a parallelogram while the member lengths stay almost unchanged. It relies strongly on joint stiffness to keep corners square. A triangle cannot change shape without at least one side changing length, so triangulation makes a frame more stable.

Diagonal bracing divides a panel into triangles and turns sideways movement into tension and compression. A truss uses straight members joined at nodes so much of the force travels axially along them. Bracing is useful when it creates a clear route; random extra pieces can increase mass without solving the weak point.

Compare designs under the same loading conditions. Ask which panel racks, which members carry force and how load reaches the supports. Looking only at the triangle count misses joint quality and unsupported compression-member length. The layout, members and connections work together.



Conceptual pin-jointed bays. The dashed outline shows a possible distortion; member and joint behaviour in your actual design must be checked.

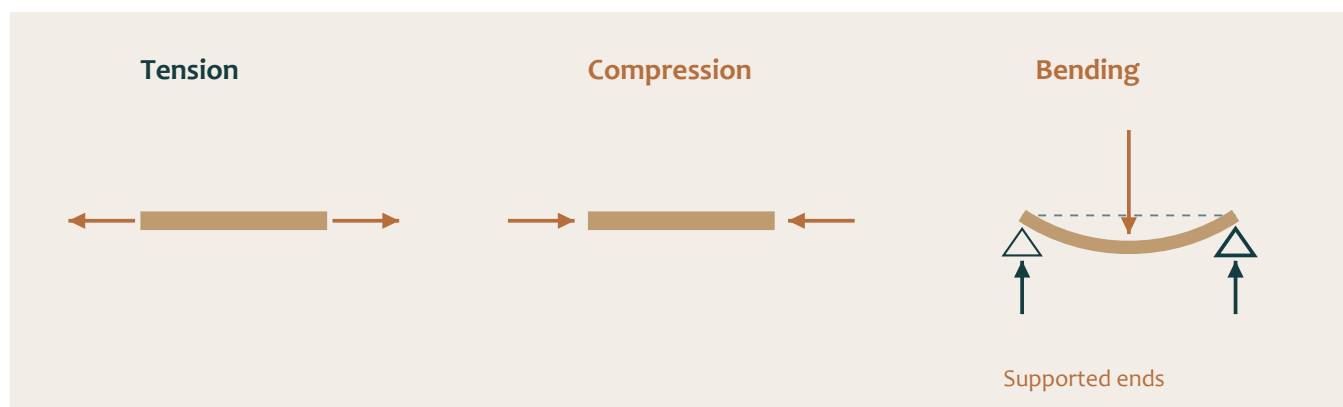
Add a diagonal to the unbraced concept panel. Use arrows and a short explanation to show how it reduces racking.

Ties pull; struts carry compression

A tie mainly carries tension: forces pull along its length. A strut mainly carries compression: forces push towards each other. Label a member by the forces it carries, not only its direction on the page. A diagonal can behave differently when loading direction or geometry changes.

Tension members need adequate cross-section and sound connections. Damage, notches or holes close to edges can concentrate stress. Compression members may bow sideways and buckle before reaching the material's crushing strength. Effective unsupported length and restraint are therefore central to the design.

Truss members are generally efficient when forces act along their axes. Bending combines compression on one side of a section and tension on another. In a simply supported beam under a central downward load, the top is compressed and the bottom is stretched. This general example explains member behaviour; it does not add concrete production to this unit.



Conceptual actions on members. Real members can experience more than one action.

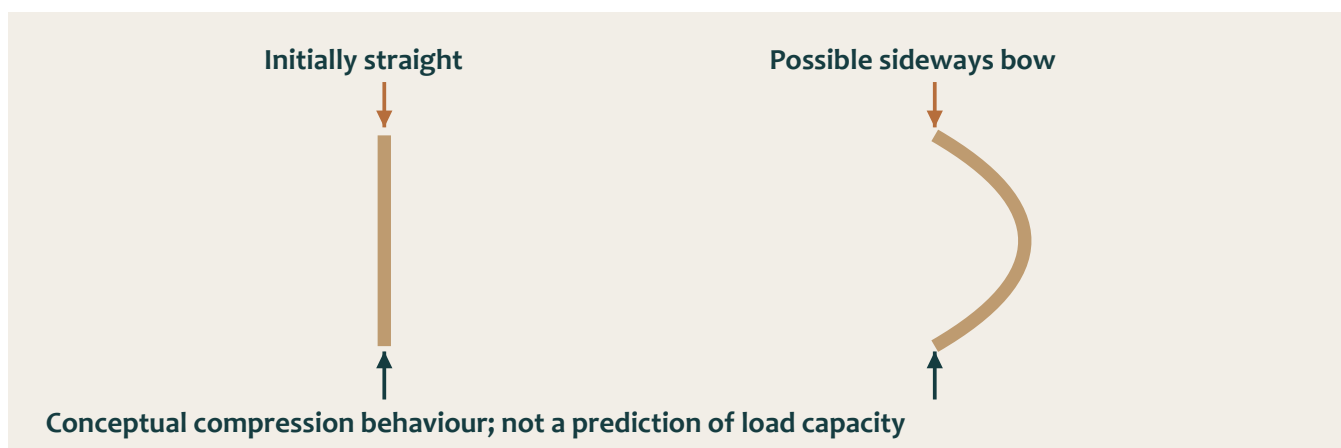
On the displayed tension, compression and bending action bars, annotate the pull, push and restraint arrows. Identify the bar in tension and the bar in compression. For the centrally loaded beam supported at both ends, label the top and bottom behaviour. Explain how the load directions produce these actions.

Why slender members buckle

Buckling is a loss of stability under compression. A long thin member can bow sideways at a load below the force needed to crush its material. A tower leg may suddenly develop a sideways bend rather than simply shortening.

Three source factors affect behaviour: slenderness, restraint and orientation. Greater unsupported length makes a member more vulnerable. Bracing at intervals can shorten effective length. Section orientation matters where a member is stiffer about one axis. The supplied square members have the specified cross-section; theory examples about other sections do not change that brief.

Use a failure observation to ask a better design question. If a member bows between two nodes, examine its length, straightness, connections and bracing. Restraint in the right place may be more useful than extra mass elsewhere. The test supplies evidence about which member or joint is critical.



Conceptual compression behaviour: a slender member may bow sideways. The diagram does not specify a failure load.

On the long-strut concept, mark a possible brace position. Explain what changes and why it could reduce buckling risk.

Name an observation to collect during the approved test that would help explain buckling.

Place material where it works

Strength and stiffness are related but different. Strength describes the load carried before failure; stiffness describes deflection under load. A frame can survive a load while still moving too much, so an evaluation should discuss both capacity and movement.

In bending, material away from the neutral axis does more work resisting tension and compression. Hollow SHS and RHS can therefore be efficient in full-size structures: material is placed around the outside rather than concentrated near the centre. These are comparisons; your model still uses the supplied 4 mm square plywood members.

Forms carry load in different ways

Truss	Triangular framework; members mainly carry axial tension or compression
Beam	Spans between supports; bending creates tension and compression
Arch	Carries compression and develops outward thrust at supports
Cantilever	Fixed at one end, free at the other; support resists bending

Choose a form by how it suits the load, span and support conditions. In the tower, the top support transfers load and the frame carries it downwards. A member or joint forced into large bending can become critical even when another part is heavily braced.

Choose two forms from the table. Explain an important difference in their load paths or supports.

Investigate, sketch and explain

A useful investigation records a pattern and explains it. Compare structures by form, material and load path rather than appearance alone. When your teacher demonstrates a simulator, keep loading conditions consistent and note what moves or fails first. Transfer the structural idea, then test the actual tower safely.

Bridge and tower investigation

Classify examples by form (beam, arch, truss, cantilever) and material; note why each form suits the span and load.

Use the PBS Bridge Basics/Challenge simulators to trial designs; capture which truss layouts reduced deflection or failed early.

Complete force practice: Forces practice (set 3).

Example / form	Material	Load path / observation	Why it suits the task

Sketching and drafting

Produce orthogonal sketches (top/front) of tower layouts showing members and joints.

Annotate load paths, reactions, likely buckling members (compression) and ties (tension).

Worksheet practice: Forces worksheet (set 1) and Forces worksheet (set 2).

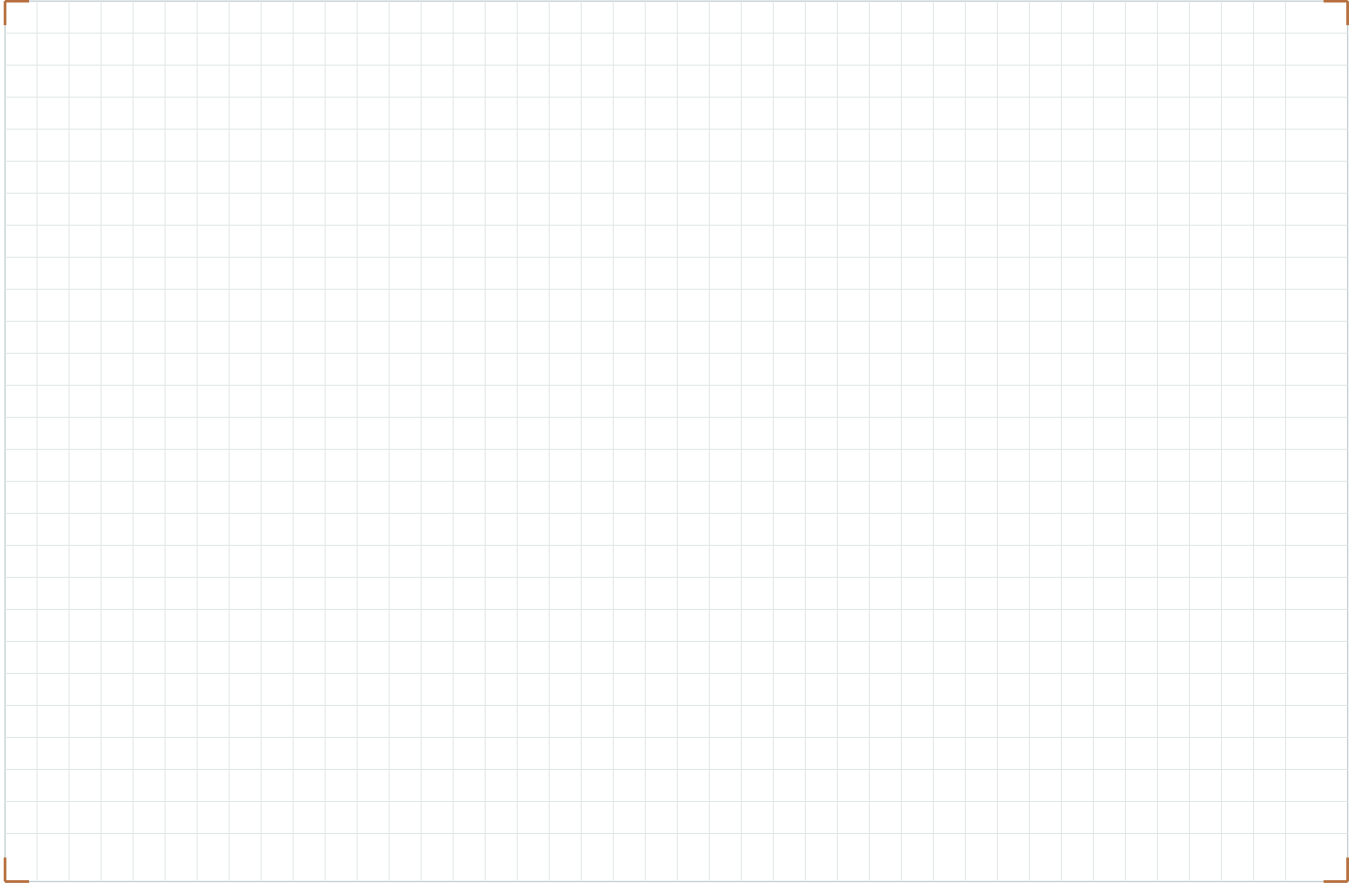
Complete the tower sketch and annotations in the working area. Linked force worksheets provide optional teacher-supplied practice; use their unchanged diagrams when selected.

Google Classroom check-in

Submit responses for Forces & Loads before starting the bridge simulator.

Capture notes in the Structure Theory & Projects task to use in your folio.

Investigate, sketch and explain - continued



Efficiency: compare like with like

A light structure is efficient only if it can safely carry the required load. The original folio divides supported load in kilograms by tower mass in grams: keep those units visible. Toolkit 2 also compares force in newtons with mass in kilograms. These conventions give different numbers; do not compare them as though their units were identical.

Worked brief-boundary calculation — not a test result

Using only the brief limits: $25\text{ kg} \div 60\text{ g} = 0.417\text{ kg/g}$ (approximately). Converting tower mass first gives $25\text{ kg} \div 0.060\text{ kg} = 416.7$ as a same-unit load-mass ratio. Using 9.8 m/s^2 gives $245\text{ N} \div 0.060\text{ kg} = 4083\text{ N/kg}$ (approximately). These describe the same hypothetical boundary in different units. Your actual ratio requires your measured result.

Costing and scaling — source comparison task

Create a spreadsheet to cost a 30 m SHS tower (150x150x8 mm @ \$246/length). Calculate total length, mass and cost.

Compare wall thickness options; discuss strength vs weight vs cost and choose the best strength-to-weight option.

The SHS task is a full-size comparison, not the model’s material brief. Use your layout to find total length; ask the teacher for stock length and the applicable mass table. Calculate mass from length $\times$ kg/m and cost from required stock lengths. The source’s \$246 per length is an exercise value, not a current verified quote.

Scaling geometry does not prove full-size capacity. Self-weight, buckling, joints and load distribution change with scale. Use the comparison to reason about material placement, mass and cost; keep engineering verification distinct from the arithmetic estimate.

Section option	Total length	Supplied kg/m	Mass	Stock lengths / cost

Structural check: loads and units

What is the difference between dead load and live load on your water tower?

Why do we convert kilograms to newtons instead of just working in kg?

What is the practical difference between a point load and a distributed load in our tests?

How do supports share the load on a simple symmetrical tower?

Structural check: shape and members

What makes triangles so effective in trusses compared with rectangles?

Why are long, slender compression members more prone to buckling?

Why do engineers prefer to keep truss members in tension or compression rather than bending?

Why can a hollow section be stronger for its weight than a solid bar?

Structural check: evidence and redesign

What information should you record during water tower load tests?

How can test results help you improve your next design?

How do you explain a failure mode using load paths and member roles?

Turn the brief into design decisions

A design brief describes what a successful tower must do. Criteria are the features you judge: height, mass, supported load and hold time. Constraints limit the choices you can make, such as the supplied member material. Read both before drawing. A light structure that carries a large load still misses the brief if it is too short or cannot support the test board.

Translate each requirement into a check you can perform. For example, show the overall height on the front view, reserve a suitable support arrangement for the test board, and plan to measure the completed tower's mass. Do not replace the issued requirements with the smaller loads, other materials or example footprint mentioned in general design notes.

Design Brief Analysis

Read and annotate the design brief, highlighting all requirements and constraints.

Create a checklist of requirements to ensure your design meets all criteria.

Requirement	Design decision / evidence to check it
At least 300 mm tall	
No more than 60 g	
At least 25 kg for at least 5 seconds	
Supports a 300 mm square board during testing	
Supplied 4 mm square cross-section plywood members	

Introduce the project goal and identify the evidence this portfolio will show.

Explain the challenge requirements and the success criteria your team had to meet.

Why triangulate every side of the tower?

Compare ideas before choosing

Generate alternatives before committing to one arrangement. A useful alternative changes something structural: bracing pattern, the number of bays, the path from the deck to the legs, or how joints meet. Compare every option against the same criteria. Appearance alone will not explain why a tower should carry load efficiently.

More members add mass and more joints to make. Fewer members may leave a long compression strut unbraced. Your choice needs a reasoned balance: a direct load path, triangulated panels, buildable joints and material used where it contributes.

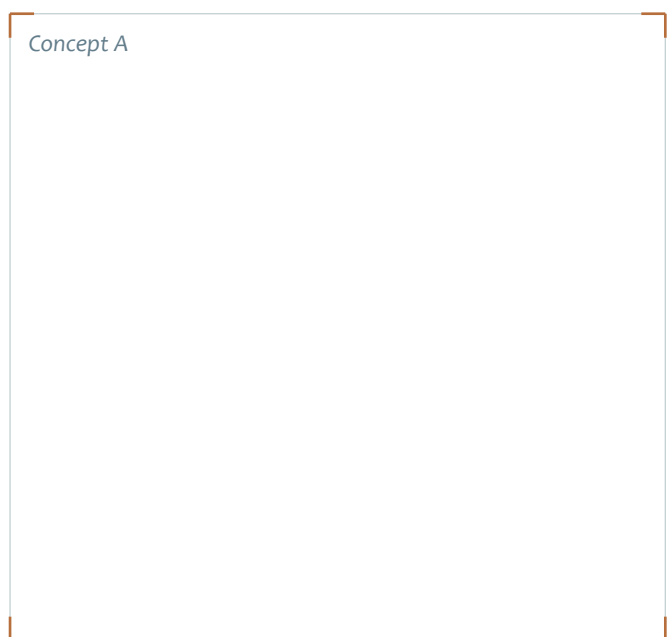
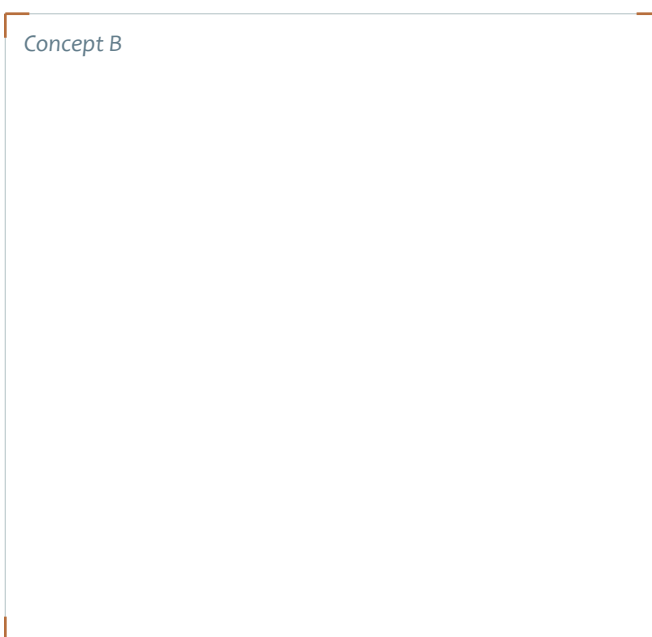
Preliminary Sketches

Sketch 2-3 different tower layout options (different bracing patterns, member arrangements).

Compare options: which has better load paths? Which is lighter? Which is easier to build?

Select your preferred option and justify your choice.

Concept A • Concept B • Concept C (optional third idea). Label the bracing and load path on each concept.

Concept A	Concept B
	

Compare load paths, likely mass and ease of assembly. State one advantage and one limitation of each idea, then justify your selected option.

Your initial design, explained

An annotated sketch records the idea you intend to investigate. Make the tower's form visible, label the deck, legs, bracing and base, then show how the board's load reaches the supports. Use arrows for the load path and a leader line from each explanation to the feature it describes. A useful note connects a feature to a purpose.

Initial Design Sketch & Annotations

Attach your initial design idea. Label key parts and explain why you chose this design.



Choose two important features in your drawing and explain how they help stability, stiffness or efficient use of mass.

Keep this initial idea as evidence. Later changes belong on the revised and final drawings so your design development remains visible.

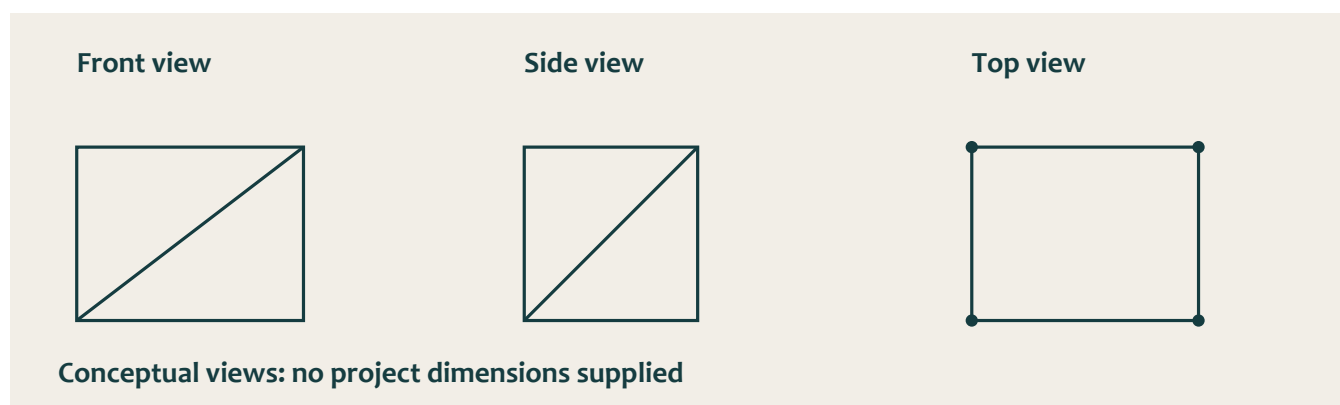
Read a design through different views

A three-dimensional sketch helps someone picture the tower, but it can hide the measurements needed to build it. Orthographic views show the object from fixed directions. The front view shows height and width; the top view shows width and depth. A side view can reveal a different bracing arrangement. Features that belong to the same part must agree between views.

Start with a clear outline and a shared orientation. Label which face is the front. Show the leg positions on the top view and align them with the corresponding legs on the front view. Add member and joint details where a small overall drawing cannot explain the connection. A dimension is the intended size; it is more dependable than measuring a printed picture.

A scale tells the reader how drawn size compares with real size. At 1:2, a feature drawn 40 mm long represents 80 mm; the written dimension remains 80 mm. This is a drawing example, not a tower cut length. State the scale on your own drawing, or write 'not to scale' if the sketch is only a concept. Do not measure from a resized reproduction to set out the project.

A view checks another view



Conceptual view conventions, not a water-tower plan. Related views must describe the same design and use written dimensions.

On the concept figure, identify a feature visible in both the front and top views. Explain one error that you could detect by comparing the views.

For an illustrative drawing at 1:2, how long would a real 120 mm feature be on paper? Show your calculation and name the dimension that should be written beside it.

Dimension the design you will build

Sketch front/top views with member lengths, base footprint, and bracing pattern. Keep bays consistent to avoid twists.

Triangulate every panel; avoid long unbraced columns. Add cross-bracing near the base where moments are highest.

The tutorial describes a common design case, rather than a rule for every tower. Moment and compression distributions depend on geometry, where and how loads act, member restraint and supports. A longer unbraced compression member is more vulnerable to buckling; length alone does not mean it carries greater compressive force. Trace the actual load path before selecting bracing.

Keep weight low; shorten compression members and use tension ties where possible.

Detailed Design • drawing evidence

Create final orthogonal sketches (top/front views) with dimensions.

Annotate load paths, member roles (struts/ties), and joint details.

Front view • Top view. Write the chosen scale and dimensions, show load arrows, label member roles and draw a joint detail where needed. Use only teacher-confirmed dimensions and connections.

Front view	Top view
	

Point out the key structural ideas (such as tension/compression) visible in your design.

Summarize your blueprint decisions, including key measurements and why they mattered.

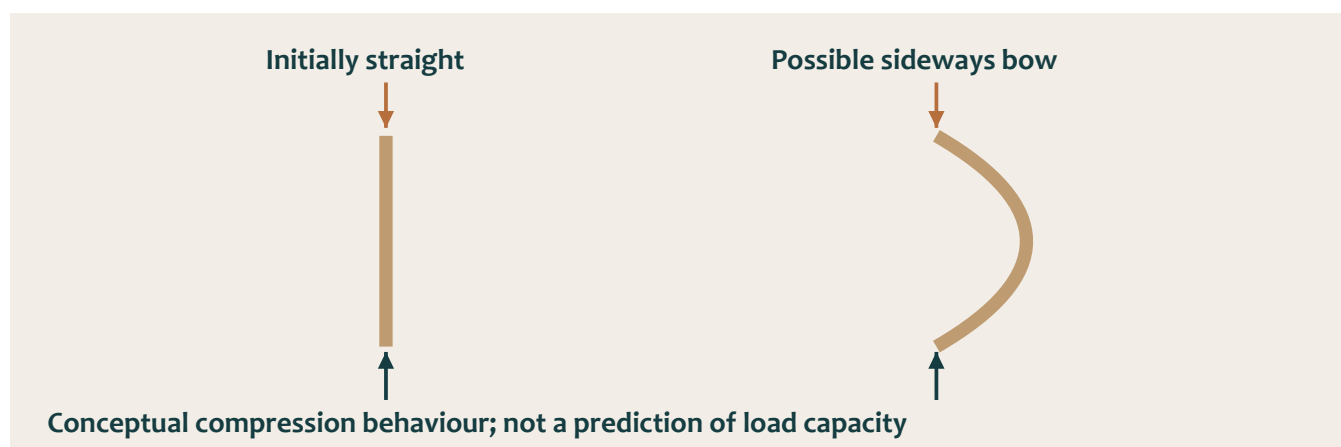
Plan members and compression

A cutting list turns the drawing into a material plan. Give each member a useful ID, then record its role, finished length, quantity and material. Match repeated pieces to the same drawing dimension. Work from an agreed datum when marking so small errors do not accumulate from one piece to the next.

The supplied tower members are 4 mm square cross-section plywood. Material comparisons in the theory help explain strength-to-weight, consistency and workability, but they do not authorise substitution. Ask the teacher to confirm permitted joint material and adhesive. Include their mass in planning as directed; the completed tower must meet the 60 g limit.

Illustrative list entry: L1 / compression leg / length from the approved drawing / required quantity / supplied member material. The missing length is deliberate: there is no standard tower cut length in this workbook. Your own drawings supply it. Before cutting, check that all four sides can be assembled from the listed pieces.

Where do you expect the highest compression in the tower?



Compression and moment distribution depend on loading, geometry and supports. A longer unbraced member is more vulnerable to buckling; length alone does not mean a larger compression force.

Record the cutting list

Create a cutting list (member lengths, quantities, materials).

Member ID / role	Finished length (mm)	Quantity	Material / joint allowance to confirm

Describe the materials and explain why they were suitable for your tower design.

Total planned member length: _____ m. Estimated material mass: _____ g. Other mass allowed by the teacher: _____ g. What will you measure before testing?

Use your layout and load position to reason about member forces. The source's expected location is a simplified prediction: actual compression and moments depend on geometry, loading, restraint and supports. Greater unbraced length increases buckling vulnerability; it does not by itself set the compression force. Explain which members carry the load and which may buckle in your design.

Plan joints and the assembly sequence

Joint examples explain force transfer through contact area. Record the method approved for your tower before adopting a teaching example as a construction detail.

Use gussets or double laps rather than single-point glue. Spread load across the joint area.

Keep opposing faces parallel; avoid gaps that cause racking. Dry-fit first, then glue.

Stage assembly symmetrically (build pairs) to control twist and keep plumb.

Adhesives need suitable surfaces, contact and curing time. Mechanical fasteners can offer removability; welding and soldering join materials in other engineering contexts. These are comparisons of methods, not an instruction to fit bolts or weld this model. Record the method approved for your tower and show its glue area, overlap or other relevant detail.

Sequence matters because a later member can block access to an earlier joint. Plan checks and curing between dependent steps. The exact order can vary with your design; explain any change rather than copying an order that cannot work.

Write a build sequence (step-by-step plan).

Teacher-confirmed joint / adhesive: _____ . Sketch its contact area in your design detail and explain which force it must transfer.

Step / operation	Quality or safety check before the next step
1.	
2.	
3.	
4.	
5.	
6.	
7.	

Resolve feedback before building

A design check gives you a chance to find weak details before material is cut. Present the drawings, cutting list and build sequence together. If feedback changes a member, joint or dimension, update all related records. A drawing and cutting list that describe different versions can produce a poorly fitting frame.

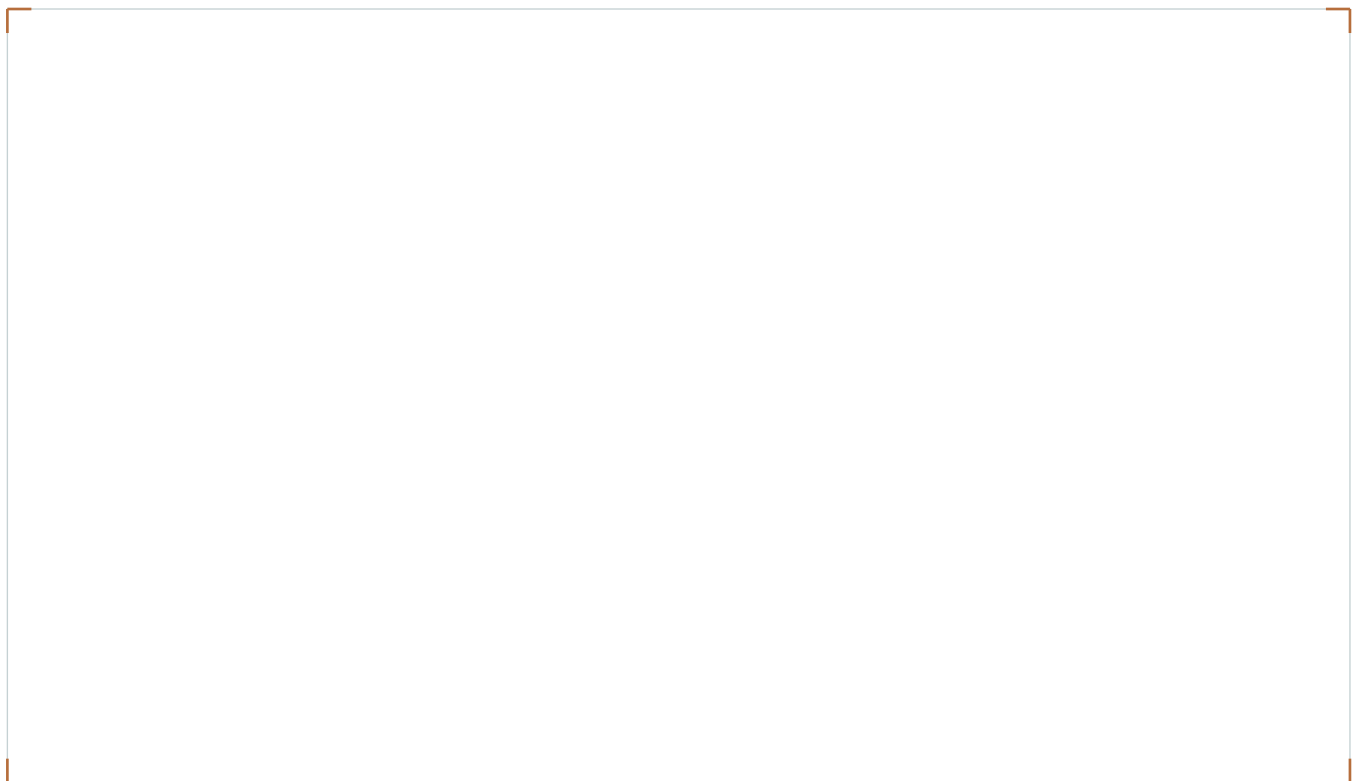
Google Classroom Check-in

Submit your design sketches and build plan before starting construction.

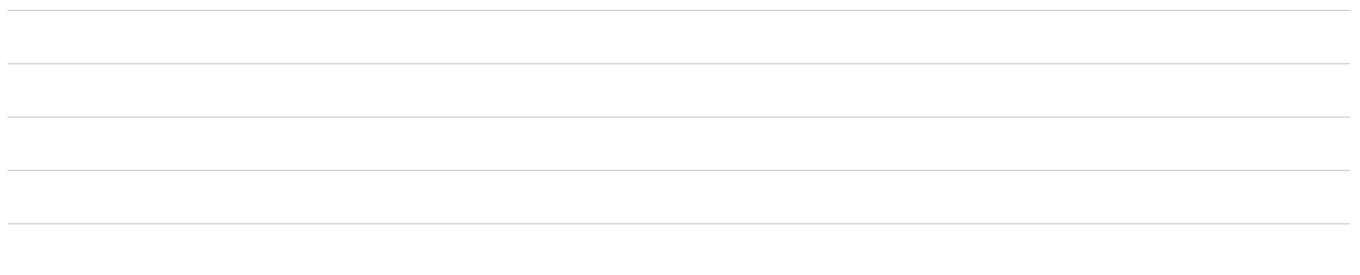
Get teacher feedback on your design before building.

Use this paper record to prepare the check-in. Your teacher directs any online submission; this workbook does not submit work or grant permission to use equipment.

Revised design / detail after feedback. Label each change and its reason. If no change is needed, explain how the design meets the criteria.



Feedback received: _____. **Action taken:** _____. **Drawing/list version:** _____. **Teacher check/date:** _____.



Make the build log useful

A build log records how the design became a real structure. Write entries as work happens so measurements and decisions do not depend on memory. Include the date, stage, people involved, work completed and the evidence reference. Record the checks that changed what you did, not just the names of tools you used.

Compare 'We glued the side' with this hypothetical entry: 'We dry-fitted the first side and checked the members against the drawing. One brace left a gap, so we stopped and checked the marked length before asking the teacher about a replacement. We rechecked the fit before bonding.' The second account links a quality problem to a decision. It does not claim that a particular tool or adhesive is approved.

A photo needs a caption to become engineering evidence. Identify the stage or member, what the image shows and why it matters. A close-up of a joint can document contact or alignment. A wide view can document the whole frame's symmetry. Keep the original file name or evidence number so a reader can locate it.

Project Photos / Video

Attach photos / videos of your project at various stages, including construction and final completion.

Rewrite 'We built the tower and it looked good' as a useful log entry. Use your own real work, or label your response as a hypothetical example.

Photos: build stages, joints, final build, test setup, failure close-up.

Plan three evidence images: a key build stage, a joint detail and the completed tower. For each, state the feature you need to show.

Record what you made and changed

Use factual language. A correction is useful evidence when you explain the cause and how you checked the result. Refer to the drawing version and evidence numbers. If the build differs from the plan, record the difference now so the evaluation can explain its effect.

Record the construction process in sequence and include one important safety choice.

Describe a build problem and clearly explain the fix your team used.

Build entry 1 • Date / stage: _____. People / contribution: _____. Work completed: _____. Safety and quality checks: _____. Issue / correction / next step: _____. Evidence reference: _____.

Build entry 2 • Date / stage: _____. People / contribution: _____. Work completed: _____. Safety and quality checks: _____. Issue / correction / next step: _____. Evidence reference: _____.

Attach or sketch two key stage images. Caption each with the stage, member/joint shown and the claim it supports. List video file / time reference if used.

How do you minimise glue joint failures?

Inspect the structure you actually built

Build quality changes structural behaviour. A twisted frame or a member that meets away from its intended node can introduce bending into a design intended to carry mainly axial forces. A gap in a bonded joint reduces contact. Check the actual tower against the drawing before the test; a good-looking outline is not enough.

Compare actual dimensions and alignment with the drawing. Check for loose joints, damage and missing bracing before testing, then record any correction so your evaluation can distinguish construction defects from design limits.

Pre-cut identical members in batches; label parts; keep offcuts for spares.

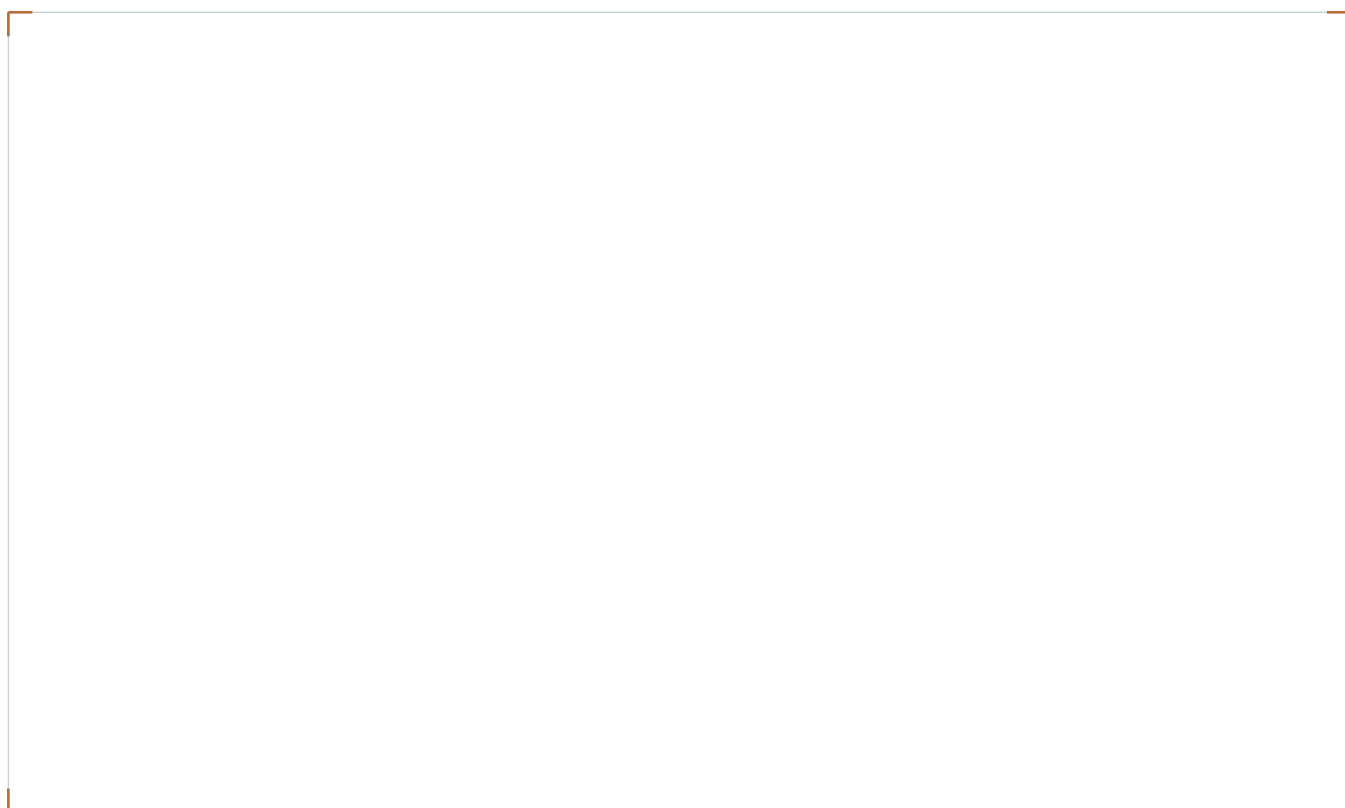
Measure base diagonals; adjust before glue cures so the frame sits flat.

Complete the inspection record. Describe evidence, not a tick based only on appearance. Refer any concern to the teacher before loading.

Check	Observed condition / measurement	Action / recheck
Height and base agree with drawing		
Frame sits flat; diagonals checked		
Legs straight and untwisted		
Bracing present and aligned at nodes		
Joints complete, sound and cured as directed		
Test board support and load point confirmed		
Mass measured under teacher's accounting rule		

Locate the inspection concern

Sketch the actual member or joint that concerns you and mark its location. Identify the concern most likely to affect your tower's performance. Predict how it could change the load path or failure mode.



Plan a controlled, safe test

A useful test changes the load in a controlled way while the other conditions stay consistent. State where the load acts, how it is applied, how long it is held and where deflection is read. A moving ruler or off-centre board can make two results hard to compare. Use a fixed reference point and record the same measurement each time.

The teacher confirms the rig, permitted increments, PPE, supervision and stop procedure. A question about near-failure testing never permits continuing beyond the approved stop condition.

The guide uses a 300 mm square board and cast iron weights, with the goal of at least 25 kg for at least 5 seconds. Bucket/loading-step examples do not replace that teacher-approved method.

Review SWMS and PPE. Clear fall zones; set clamps/fixtures and loading plate/bucket centrally.

Add load in steps; note load at first deflection and at failure. Use steady increments for clean data.

Film side and front views; mark reference points for deflection if possible.

Explain your testing method and why it was a fair way to evaluate performance.

Tower ID / date / group: _____. Load point and fixture: _____. Board and weights: _____. Approved increments: _____. Hold-time method: _____. Deflection datum / instrument: _____. PPE / fall zone / stop signal: _____. Supervisor and recording roles: _____.

Safety Prep • source checklist

SWMS reviewed and controls in place for tower/beam testing.

PPE on: eye protection, enclosed shoes, apron/long sleeves as required; hair and cords secured.

Work area clear; clamps/fixtures set; fall zone clear; first aid kit accessible; supervisor aware.

Load path safe: bucket/plate centred, no hands under load, clear stop protocol if instability occurs.

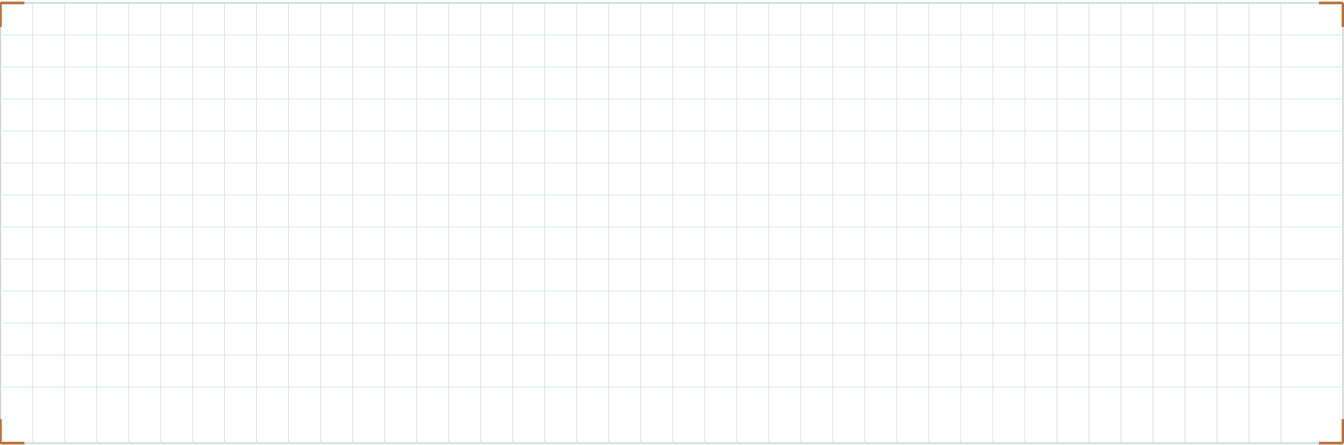
Turn the test into evidence

Record the total load after each teacher-directed step, not just the added amount. State whether deflection was measured or estimated. Record the first visible movement and the first failure separately from the final load. If the test is stopped safely before failure, say so; do not invent a failure load.

Tower ID: _____. Height: _____ mm. Tower mass: _____ g. Board / fixture mass accounting: _____. Test method reference: _____. Hold time at target load: _____ s.

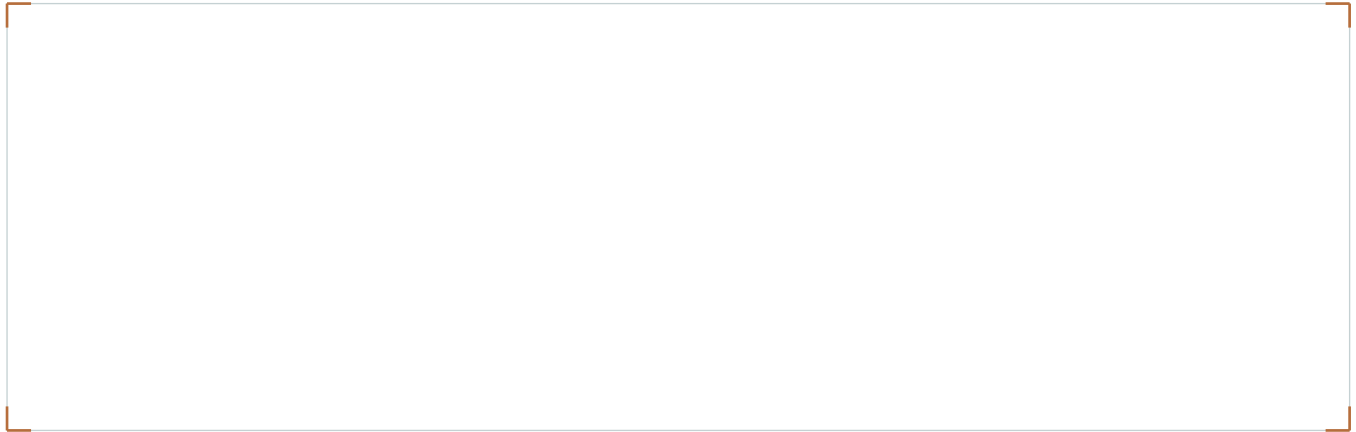
Step	Total load (kg)	Hold time (s)	Deflection / lean (mm)	Observation / evidence
1				
2				
3				
4				
5				
6				

Plot load against deflection using your recorded values. Label the scale and units. Show a measured trend; mark estimates clearly.

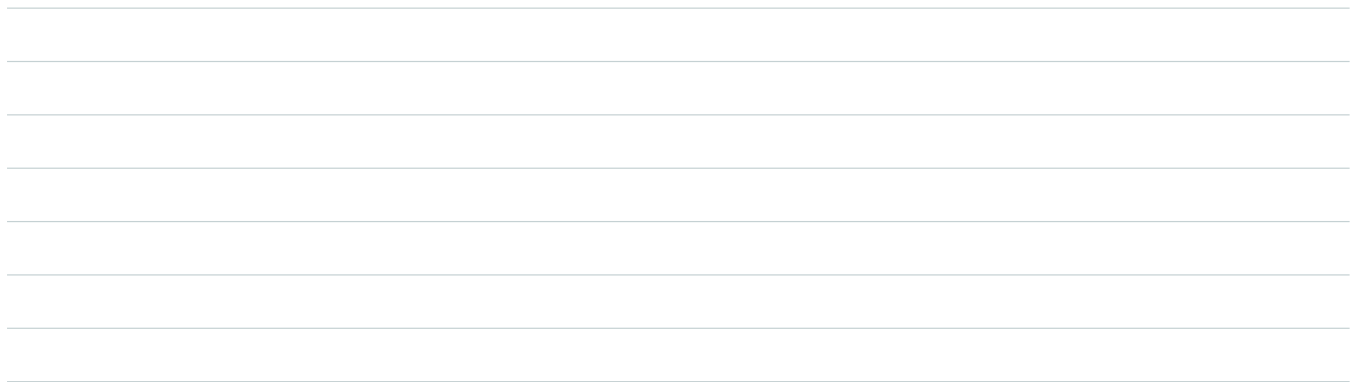


Locate and interpret the first failure

First deflection / load: _____. Peak recorded load: _____. First failure member or joint: _____. Failure mode and observation: _____. Photo/video reference: _____. Sketch or label the observed failure location below; keep this observation separate from your explanation.



Report your measured results and interpret what they say about tower performance.



Measurement & Data • source checklist

Record tower/beam mass before testing.

Capture load increments and total load at failure (note timing of first deflection).

Note deflection/lean at each major increment (estimate or measure at marked reference points).

Identify first failure point: member buckling, joint shear/peel, concrete crack, tension break.

Compare efficiency with units intact

A tower that carries the largest load is not necessarily the most efficient use of material. A load-to-mass ratio compares supported load with tower mass. The original folio expresses supported load in kilograms and tower mass in grams. Keep those units attached when reporting the result so a ratio calculated with kilograms in the denominator is not confused with it.

Illustrative worked example

Hypothetical tower A holds 20 kg and has a mass of 50 g. Its folio ratio is $20 \div 50 = 0.40 \text{ kg/g}$. Tower B holds 18 kg and has a mass of 40 g: $18 \div 40 = 0.45 \text{ kg/g}$. A holds more load; B has the higher load-to-mass ratio. These invented values explain the calculation and are not student test results or approved targets.

Module 4 also describes load force in newtons divided by tower mass in kilograms. Using the classroom approximation 10 N per kg, tower A's 20 kg load is about 200 N and its 50 g mass is 0.050 kg: $200 \div 0.050 = 4000 \text{ N/kg}$. This is a differently expressed quantity. Label the formula and units whenever comparing groups.

Load to Tower Mass Ratio (complete this after we test our towers)

Calculate and record your load-to-mass ratio using the formula:

Load to Mass Ratio = Load Supported (kg) / Tower Mass (g)

Load Supported: _____ kg Tower Mass: _____ g Load to Mass Ratio: _____

Use your actual result and one teacher-provided comparison result. Which has the greater supported load and which has the greater kg/g ratio? Explain why these judgements can differ.

What should you record during load testing?

Data: mass of tower/beam, load steps, peak load, estimated deflection, material list.

Judge the result using evidence

Evaluation makes a judgement from evidence. Identify what happened, connect it to a structural idea and explain why it matters. A first failure near a joint is an observation; a claim that poor contact caused it needs evidence such as a visible gap. State uncertainty when the evidence does not settle the cause.

PMI helps you examine the result: Plus records successful features, Minus records limitations, and Interesting records useful observations or questions. Follow it with a specific improvement. Changing the feature most closely linked to first failure or excessive deflection gives a clearer comparison than changing everything at once.

Analyze where failure happened and support your explanation with exact data.

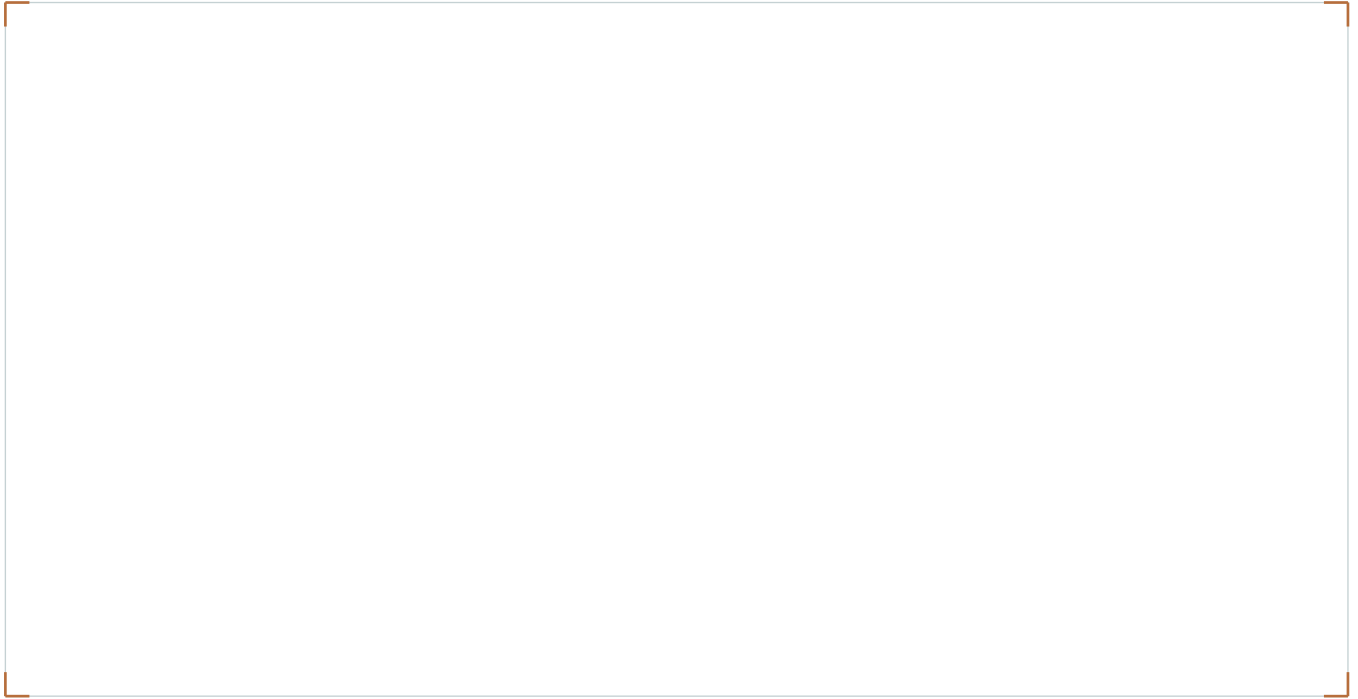
Complete the PMI table with evidence from your design, build and test. Make each entry specific enough to support an evaluation.

Plus • what worked; evidence	Minus • limitation; evidence	Interesting • question / unexpected finding

Redesign and practical reflection

Propose a realistic redesign change and explain how it would improve outcomes.

Sketch one proposed change. Label the feature, expected effect on load path/stiffness/efficiency and what a repeat test would compare.



Name one safety control actually used and explain its purpose. Then explain one material/waste or maintenance decision relevant to the model or a future full-size tower.

Reflection: what failed first, why (buckling, joint peel, shear), one change to improve stiffness/weight/corrosion protection.

Explain the test: model, strength and failure

These questions retain the wording of the source Water Tower Test & Folio Q&A. Explain the engineering ideas in your own words. Use an actual observation where relevant; label any illustrative example.

Why do we build and test a small model water tower instead of just calculating a full-size 30 m design?

What is the difference between strength and stiffness in your water tower?

Why do we test towers close to failure instead of stopping when they "look safe"?

Testing follows the teacher-approved method and stop conditions. A question about failure does not authorise a test.

Explain the test: control, quality and error

Use the test and inspection records to explain why data may differ between groups. The source's reference to beam tests is retained in Question 6; answer it for the Water Tower work in this unit.

What makes a good test method for your water tower?

Why might two towers with almost the same design fail at very different loads?

What are the main sources of error in your tower and beam tests?

Explain the evidence and the redesign

A clear folio links a claim to a record. Use the graph, test table, inspection notes and photographs to support these answers.

How can graphs and tables improve your folio?

What makes a strong evaluation paragraph in your folio?

How should you talk about safety and environment in your evaluation?

When you re-design, should you change everything at once?

Original folio: decisions and structural ideas

These are the original Water Tower Folio learning prompts. Use your own project evidence. The source folio's historical /65 and question marks do not apply to the current ungraded formative learning task.

The Design Process

Describe the design process you followed for this project. Outline the key steps from initial brainstorming to final construction.

Project Limitations

Discuss the limitations and constraints that affected your design choices, such as material restrictions, weight limits, height requirements, and time constraints.

Short Answer Questions (2 - 3 sentences)

Why is structural stability important in the design of a water tower?

How does the choice of materials affect the strength of the tower?

What forces act on your tower when supporting a load, and how did your design counteract them?

Original folio: improvement and teamwork

Short Answer Questions (2 - 3 sentences)

Explain a challenge your group faced during construction and how you solved it.

How could you improve your tower design if given more time and resources?

Why is weight distribution important when designing a structure that supports a heavy load?

Group Performance Evaluation

Evaluate each group member’s performance, including their contribution to the design, construction, and teamwork.

Describe contributions fairly and specifically. The five source rows provide response space; your teacher confirms the group size. Leave unused rows blank.

Group Member	Strengths	Areas for Improvement
1)		
2)		
3)		
4)		
5)		

Original long response 1: predict and assess

Long Response Question 1

Critically examine how your project will handle different amounts of weight. Discuss any potential weaknesses and assess how well your design choices help to prevent or reduce these issues.

Your response should be well-structured and supported with reasoning based on your design, materials, and construction process.

Use 4–8 paragraphs as the original folio directs. Examine load paths, long compression members, joints and the effect of uneven loading. Support predictions with your design and the evidence you have; distinguish a predicted weakness from an observed failure. Continue on page 58 or a labelled attached sheet if needed.

Using the ALARM Matrix

A (Acknowledge): Identify the main idea or issue.

L (Link): Connect the idea to relevant knowledge or concepts.

A (Analyze): Break down the key components and their impact.

R (Relate): Compare with examples, experiences, or similar designs.

M (Make a Judgment): Conclude with an evaluation of your design's likely performance.

Write your response below. Refer to a labelled feature or evidence record in each main paragraph.

Original long response 2: evaluate the test

Long Response Question 2

Evaluate how effectively your tower design supported the weight during testing. What design features contributed to its success or failure, and how would you modify your design to improve its performance in future builds?

Your response should be well-structured and supported with reasoning based on your design, materials, and construction process.

Use 4–8 paragraphs, with an introduction, developed body and conclusion. Judge how well the tower met the brief; explain successful and weak features through the measured test; then justify specific changes. Use past tense for what happened and conditional language for proposed improvements. Continue on page 58 or a labelled attached sheet if needed.

The essay guide asks you to use facts, discuss shape/materials/joints and explain what you would change and why. Link cause and effect clearly: 'because', 'therefore' and 'as a result' connect an observation to reasoning. Do not replace test evidence with a general statement that the tower was good.

Write your response below. Check that it answers performance, contributing features and future modifications.

Show the final design and its development

Final Design Sketch & Annotations

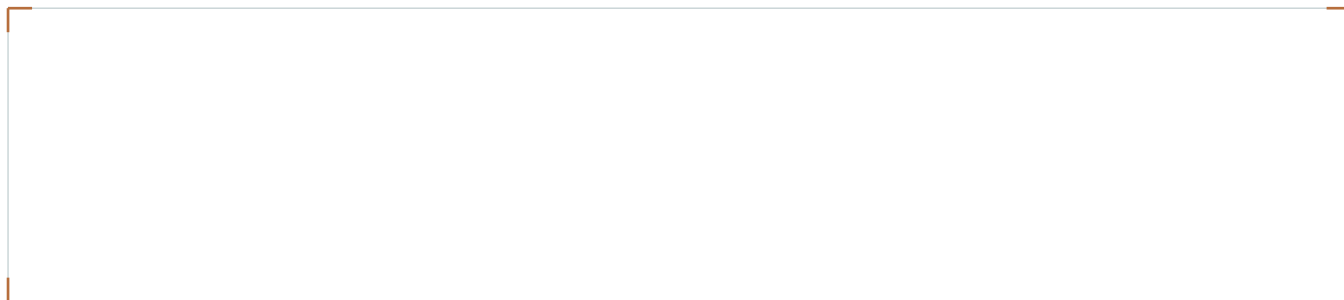
Attach or draw the final version of your design. Label key features and explain any modifications made from the initial concept.



Represent the final version accurately. Use member labels that match your test and build records. Show changes from the initial sketch, explain the reason and identify whether the change was made before or after testing. A proposed future redesign should be labelled separately so it is not mistaken for the tested tower.

Show the final design and its development - continued

Drawing/attachment field. Include labelled structural features, relevant dimensions and a clear load path. Refer to page 31 for the dimensioned views rather than crowding this sketch.



Modification • reason • when made • evidence reference. If the final version matches the initial idea, justify retaining it using the evidence.

Develop the judgement with evidence

Use the scaffold to strengthen a long response, then continue the response that needs more space. Give the question number and refer to the earlier page. A useful evaluation shows what you observed, what it suggests, and which evidence limits the judgement.

Introduction

What was the aim of the project?

What materials did you use to build your tower?

What were you trying to test?

Design (What worked well?)

What parts of your design were strong?

What helped your tower hold weight? (e.g. shape, structure, joints)

Limitations (What didn't work well?)

Did any part of your tower bend, break, or fall?

Why do you think that happened?

Conclusion (How could you improve it?)

What changes would you make next time?

How would you make your tower stronger?

Essay self-check

Did I answer all parts of the question?

☐ Checked

Do I have an introduction, body, and conclusion?

☐ Checked

Did I explain my ideas clearly?

☐ Checked

Did I check for spelling and punctuation?

☐ Checked

Name the next revision needed and the sentence, paragraph or evidence it improves. If no revision is needed, give one specific reason based on the four checks.

Continuation of Long Response ____ from page _____. Use evidence references for claims and finish with a justified judgement.

READ / INVESTIGATE / APPLY

Make the evidence easy to find

A folio is an organised engineering account. Give drawings, photos, tables and graphs a label, then use that label when explaining a decision. Check that every claim points to genuine work or measured data. An unfinished requirement is recorded as pending with its next action.

Complete the evidence index with the location of each record and any next action. Include additional long-response sheets.

Evidence	Page / attachment / file reference	Complete / next action
Brief and criteria; sketches and design feedback		
Materials/tools list and sequence		
WHS hazards/controls, PPE and SWMS		
Dated build log and stage images		
Test setup, mass, load, time and observations		
Load–deflection table/graph and failure evidence		
PMI, original responses and justified improvement		
Final drawing, attachments and learning reflection		

Evidence Capture • source checklist

Photos: build stages, joints, final assembly, test setup, failure close-up.

Video: side and front angles during loading, showing deflection and failure.

Costing/scale sheet updated (tower) and material list saved.

Reflection drafted: what failed first, why, and one change to improve stiffness/weight/corrosion protection.

Capture your key engineering learning and how you will apply it in future work.

Teacher to confirm the group size, delivery date, approved practical method and any full-size scale/costing extension. The original folio marks are historical; the current task is formative.